

Kinora Cervical Spine

Clinical AI Orientation (Part 12)

Vertebrae, Joints, Ligaments, Muscles, Nerves, Biomechanics, Pathologies & Rehabilitation

Version 1.0 — Kinora Admin Conocimientos Upload

Disclaimer

*Educational Kinora AI resource — not clinical care. Sources: Gray's/Standring, Moore, Bogduk, Neumann, Magee, APTA Neck Pain CPG themes. **Red flags** — cervical myelopathy, progressive neuro deficit, suspected fracture/instability (Canadian C-Spine/NEXUS), cervical arterial dissection features (sudden severe neck/head pain, Horner, cranial nerve signs), inflammatory atlantoaxial instability — require urgent medical referral. Screen for cervical arterial dysfunction per current guidelines before end-range manual techniques.*

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1. Vertebrae

RECORD: Atlas (C1)

ID: CERV-VERT-01

Vertebra: Atlas (C1)

Type: Atypical — no body/spinous process; anterior & posterior arches + lateral masses

Landmarks: Anterior tubercle, posterior tubercle, superior articular facets (concave for occipital condyles), inferior facets, transverse foramen, groove for vertebral artery on posterior arch

Muscle Attachments: Rectus capitis anterior/lateralis; obliquus capitis superior/inferior; levator scapulae; splenius/scalenes contributions via TP

Ligament Attachments: Transverse ligament of atlas (holds dens); alar ligaments (occiput–dens); anterior/posterior atlanto-occipital membranes; facet capsules

Blood Supply: Vertebral arteries through transverse foramina; anastomoses around OA joint

Biomechanics: Atlanto-occipital: primarily flexion/extension (nodding ~10–25° combined teaching ranges). Atlas rotates on axis via dens pivot.

Clinical Importance: Jefferson fracture (burst); transverse ligament integrity critical (ADI); vertebral artery groove trauma risk; rheumatoid atlantoaxial instability screening

Imaging: Open-mouth odontoid; lateral ADI; CT trauma; MRI ligaments/cord

Notes: ADI adults normally ≤3 mm; increased ADI suggests transverse ligament incompetence

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Axis (C2)

ID: CERV-VERT-02

Vertebra: Axis (C2)

Type: Atypical — dens (odontoid process); bifid SP often large

Landmarks: Dens, superior articular facets (nearly flat for atlas), inferior facets (typical orientation begins), pars/pedicle, transverse foramen, bifid SP

Muscle Attachments: Obliquus capitis inferior; rectus capitis posterior major; multifidus/semispinalis; levator scapulae

Ligament Attachments: Alar ligaments; apical ligament; transverse ligament articulates dens posteriorly; tectorial membrane; facet capsules

Blood Supply: Vertebral arteries; ascending cervical contributions

Biomechanics: Atlantoaxial complex provides ~50% of cervical rotation (dens pivot). Flexion/extension less than OA.

Clinical Importance: Odontoid fractures (Anderson-D'Alonzo I–III); Hangman's fracture (traumatic spondylolisthesis C2); dens nonunion risk Type II

Imaging: Open-mouth; CT dens; MRI cord/ligaments

Notes: Rule out dens injury before mobilization in trauma

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: C3 Vertebra

ID: CERV-VERT-03

Vertebra: C3 Vertebra

Type: Typical cervical

Landmarks: Small body, bifid SP, uncinate processes, transverse foramen, superior facets face posteromedially-upward

Muscle Attachments: Longus colli/capitis, scalenes, multifidus, semispinalis, splenius, levator, trapezius via SP/ligamentum nuchae continuum

Ligament Attachments: ALL, PLL, ligamentum flavum, interspinous, nuchal, uncovertebral capsules, facet capsules

Blood Supply: Vertebral artery in foramen; ascending cervical

Biomechanics: Segmental flexion/extension/sidebend/rotation with coupled motion (lower cervical typically ipsilateral SB-rotation coupling)

Clinical Importance: Disc/facet degeneration; radiculopathy less common than C5–C7; myelopathy if canal narrows

Imaging: XR lateral/AP/oblique; MRI disc/cord

Notes:

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: C4 Vertebra

ID: CERV-VERT-04

Vertebra: C4 Vertebra

Type: Typical cervical

Landmarks: Uncinate processes; bifid SP; transverse foramen

Muscle Attachments: As C3 — longus, scalenes, deep extensors, trapezius/levator

Ligament Attachments: ALL/PLL/flavum/nuchal/uncovertebral/facet

Blood Supply: Vertebral artery

Biomechanics: Mid-cervical mobility; coupled SB/rotation

Clinical Importance: Common degenerative level; referred scapular pain patterns

Imaging: MRI for radiculopathy workup

Notes:

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: C5 Vertebra

ID: CERV-VERT-05

Vertebra: C5 Vertebra

Type: Typical cervical

Landmarks: Uncinates; SP bifid; large transverse processes relative

Muscle Attachments: Scalenes, longus, multifidus, trapezius, levator, rhomboid minor near C7–T1 but attachments cascade

Ligament Attachments: Standard cervical ligaments + uncovertebral

Blood Supply: Vertebral artery; C5 root exits above C5 pedicle (cervical roots exit above same-numbered vertebra except C8)

Biomechanics: High mobility segment; C5 radiculopathy (deltoid/biceps) common

Clinical Importance: Disc herniation C4–5; facet arthropathy; foraminal stenosis

Imaging: MRI; oblique XR foramina

Notes: C5 myotome: shoulder ABD; reflex: biceps

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports

RECORD: C6 Vertebra

ID: CERV-VERT-06

Vertebra: C6 Vertebra

Type: Typical cervical — carotid tubercle (Chassaignac) on TP anterior

Landmarks: Carotid tubercle landmark; bifid SP; uncinates

Muscle Attachments: Scalenes (important), longus, deep neck extensors, trapezius

Ligament Attachments: Standard + uncovertebral prominent clinically

Blood Supply: Vertebral artery; carotid pulse palpable at Chassaignac

Biomechanics: C5–6 often most mobile lower cervical; frequent spondylosis site

Clinical Importance: C6 radiculopathy (wrist extensors/biceps); uncovertebral osteophytes → foraminal stenosis

Imaging: MRI; CT for bony stenosis

Notes: C6 myotome: wrist extension; reflex: brachioradialis

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: C7 Vertebra

ID: CERV-VERT-07

Vertebra: C7 Vertebra

Type: Transitional — vertebra prominens; often non-bifid SP; small/absent transverse foramen for vertebral a. (vein may pass)

Landmarks: Vertebra prominens SP; transitional facets toward thoracic; uncinates smaller

Muscle Attachments: Trapezius, rhomboid minor, levator, serratus posterior superior, scalenes, erectors

Ligament Attachments: Nuchal ligament ends near C7; ALL/PLL/flavum; first rib relation via C7 TP variants (cervical rib)

Blood Supply: Vertebral artery usually enters at C6; C7 foramen often only venous

Biomechanics: Less rotation than mid-cervical; load transfer to T1

Clinical Importance: C7 radiculopathy (triceps/wrist flexors/finger extensors); cervical rib/TOS; clay-shoveler's SP avulsion

Imaging: XR; MRI; rule out cervical rib on AP

Notes: C7 myotome: elbow extension; reflex: triceps

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Typical vs Atypical Summary

ID: CERV-VERT-08

Vertebra: Typical vs Atypical Summary

Type: Classification record

Landmarks: Atypical: C1 (no body/SP), C2 (dens), C7 (prominens/transitional). Typical: C3–C6 (uncinates, bifid SP, transverse foramen for VA)

Muscle Attachments: See individual records

Ligament Attachments: See individual records

Blood Supply: VA course C1–C6 typically

Biomechanics: Atypical anatomy drives unique fracture patterns and instability rules (ADI, dens, Jefferson)

Clinical Importance: Never treat C1–C2 like typical segments clinically

Imaging: Trauma imaging algorithms (Canadian C-Spine / NEXUS) then targeted CT

Notes: Canadian C-Spine Rule and NEXUS for imaging decisions after blunt trauma

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

2. Joints

RECORD: Atlanto-occipital (C0–C1)

ID: CERV-JNT-01

Joint: Atlanto-occipital (C0–C1)

Type: Synovial condyloid

Capsule: Loose capsule allowing nodding

Ligaments: Anterior/posterior atlanto-occipital membranes; OA joint capsules; alar influence

Motion: Primarily flexion/extension (nodding); minimal rotation/SB

Arthrokinematics: Occipital condyles convex on concave atlas facets — opposite roll/glide teaching

Osteokinematics: Combined OA flexion/extension often cited ~15–25°; rotation minimal

Stability: Bony congruence + membranes; alar ligaments limit extremes

Pain Referral: Suboccipital headache; occipital referral

Clinical Tests: OA flexion-extension mobility; Sharp-Purser is AA not OA; rule out VBI/cervical arterial dysfunction per guidelines before end-range

Pathologies: OA headache, rheumatoid involvement rare vs AA, fracture

Treatment: Manual therapy only after red-flag/CAD screen; deep neck flexor + suboccipital motor control

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Atlanto-axial (C1–C2)

ID: CERV-JNT-02

Joint: Atlanto-axial (C1–C2)

Type: Synovial pivot (dens) + plane facet joints

Capsule: Facet capsules; cruciform/transverse ligament complex critical

Ligaments: Transverse ligament of atlas; alar; apical; tectorial membrane; AA facet capsules

Motion: ~40–50% of cervical rotation from AA complex (teaching approximation)

Arthrokinematics: Atlas rotates about dens; lateral masses glide

Osteokinematics: Rotation primary; flexion/extension secondary smaller

Stability: Transverse ligament prevents anterior atlas subluxation on dens; alar limit rotation/SB extremes

Pain Referral: Upper cervical, suboccipital, sometimes facial

Clinical Tests: Sharp-Purser (transverse ligament — trained clinicians); lateral shear; ADI on imaging; CAD screen

Pathologies: AA instability (RA, Down syndrome, trauma); dens fracture; alar injury (whiplash)

Treatment: Stabilize/refer instability; never aggressive rotation if instability suspected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Cervical Facet (Zygapophyseal) Joints C2–3 to C7–T1

ID: CERV-JNT-03

Joint: Cervical Facet (Zygapophyseal) Joints C2–3 to C7–T1

Type: Synovial plane joints

Capsule: Fibrous capsule richly innervated (medial branches)

Ligaments: Facet capsules; ligamentum flavum anteriorly related; interspinous/supraspinous/nuchal posterior

Motion: Guide SB and rotation; bear load in extension

Arthrokinematics: Inferior facet of superior vertebra glides on superior facet of inferior vertebra

Osteokinematics: Orientation ~45° in mid-cervical — favors coupled motion

Stability: Capsule + muscular control (multifidus)

Pain Referral: Cervical facet referral maps (e.g., C2–3 → headache; C5–6 → scapular) — classic Bogduk patterns approximate

Clinical Tests: Extension-rotation provocation; palpation; medial branch blocks diagnostic (interventional)

Pathologies: Facet syndrome, arthropathy, whiplash facet pain, fracture

Treatment: Manual therapy + exercise; radiofrequency neurotomy selected chronic facet pain after dual blocks

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Uncovertebral Joints (of Luschka)

ID: CERV-JNT-04

Joint: Uncovertebral Joints (of Luschka)

Type: Pseudoarthroses / synovial clefts along uncinate processes C3–C7

Capsule: Uncinate process articulates with bevel of superior vertebral body

Ligaments: Not classic ligaments — disc/uncinate complex; osteophytes common

Motion: Guide SB; limit translation; foraminal relations

Arthrokinematics: Uncinate guides coupled motion; osteophytes encroach foramen/VA

Osteokinematics: Sidebending constrained by uncinate

Stability: Bony + disc integrity

Pain Referral: Neck pain with SB; arm symptoms if foraminal stenosis

Clinical Tests: Spurling related; imaging for uncovertebral hypertrophy

Pathologies: Uncovertebral arthrosis → foraminal stenosis/radiculopathy

Treatment: Traction trial, nerve mobility, surgery (foraminotomy) selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Cervical Intervertebral Discs

ID: CERV-JNT-05

Joint: Cervical Intervertebral Discs

Type: Fibrocartilaginous symphysis (anulus + nucleus); no disc between C0–C1 or C1–C2

Capsule: Anulus fibrosis; nucleus pulposus (less gelatinous with age); cartilaginous endplates

Ligaments: ALL/PLL continuity; uncovertebral clefts laterally

Motion: Shock absorption; height for foramina; motion segment flexibility

Arthrokinematics: Compression/shear/torsion loading; posterolateral herniation toward root

Osteokinematics: Segmental DF/PF/SB/rotation shared with facets

Stability: Anulus + PLL; posterior herniation risks cord (myelopathy) vs root (radiculopathy)

Pain Referral: Neck pain ± radicular arm pain in dermatome; cough/sneeze may worsen

Clinical Tests: Spurling, distraction, ULNT1, reflex/myotome/sensory screen

Pathologies: Disc herniation, DDD, inflammatory discitis (red flag)

Treatment: Nonsurgical care first for most radiculopathy; ACDF/disc arthroplasty selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

3. Ligaments

RECORD: Anterior Longitudinal Ligament

ID: CERV-LIG-01

Ligament: ALL

Origin: Occiput/C1 continuum down anterior bodies

Insertion: Anterior vertebral bodies/discs to sacrum

Function: Limits extension; reinforces anulus anteriorly

Clinical: Hyperextension injury; ossification (DISH) can stiffen

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Posterior Longitudinal Ligament

ID: CERV-LIG-02

Ligament: PLL

Origin: Body of axis / tectorial membrane continuum

Insertion: Posterior bodies/discs within canal

Function: Limits flexion; bars central disc somewhat — herniations often posterolateral

Clinical: OPLL → myelopathy in some populations

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Ligamentum Flavum

ID: CERV-LIG-03

Ligament: Ligamentum flavum

Origin: Lamina above

Insertion: Lamina below

Function: Elastic; limits flexion; restores extension

Clinical: Hypertrophy → stenosis/myelopathy

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Interspinous Ligament

ID: CERV-LIG-04

Ligament: Interspinous

Origin: SP above

Insertion: SP below

Function: Limits flexion

Clinical: Whiplash sprain continuum

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Supraspinous / Nuchal Ligament

ID: CERV-LIG-05

Ligament: Supraspinous continuous with ligamentum nuchae in cervical region

Origin: EOP / nuchal line

Insertion: Cervical SPs to C7 then thoracic supraspinous

Function: Limits flexion; muscle attachment septum; proprioception

Clinical: Nuchal pain; attachment for trapezius/splenius

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Alar Ligaments

ID: CERV-LIG-06

Ligament: Alar ligaments

Origin: Dens superolateral

Insertion: Occipital condyles (medial)

Function: Limit AA rotation and OA SB extremes

Clinical: Whiplash/trauma; instability if torn — specialist assessment

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Transverse Ligament of Atlas

ID: CERV-LIG-07

Ligament: Transverse ligament (cruciform complex horizontal band)

Origin: Atlas lateral mass

Insertion: Opposite lateral mass behind dens

Function: Primary restraint preventing anterior atlas subluxation on dens

Clinical: RA/Down syndrome/trauma — check ADI; Sharp-Purser by trained clinicians only

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

4. Muscles

RECORD: Longus Colli

ID: CERV-MUS-01

Muscle: Longus Colli

Origin: Cervical/thoracic bodies/TPs

Insertion: Bodies/TPs superiorly to atlas

Innervation: C2–C6 ventral rami

Function: Deep neck flexion; segmental stability

Clinical Importance: Inhibited in neck pain; DNF training target

Exercises: Supine cranio-cervical flexion (CCFT progressions)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Longus Capitis

ID: CERV-MUS-02

Muscle: Longus Capitis

Origin: C3–C6 TPs

Insertion: Occipital basilar part

Innervation: C1–C3

Function: Capital flexion; OA control

Clinical Importance: With longus colli in DNF synergy

Exercises: CCFT; chin nod holds

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Multifidus (Cervical)

ID: CERV-MUS-03

Muscle: Multifidus (Cervical)

Origin: Articular pillars/TPs

Insertion: SPs 2–4 levels above

Innervation: Dorsal rami

Function: Segmental extension/rotation control

Clinical Importance: Atrophy/fatty infiltrate in some neck pain/whiplash cohorts

Exercises: Prone/4pt activation; with DNF co-contraction

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Semispinalis Capitis/Cervicis

ID: CERV-MUS-04

Muscle: Semispinalis Capitis/Cervicis

Origin: TPs thoracic/cervical

Insertion: Occiput / cervical SPs

Innervation: Dorsal rami

Function: Extension; cervical lordosis support

Clinical Importance: Endurance deficits common in neck pain

Exercises: Prone lifts graded; avoid global shrugging

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Suboccipitals (RCPmaj/min, OCSup/Inf)

ID: CERV-MUS-05

Muscle: Suboccipitals (RCPmaj/min, OCSup/Inf)

Origin: C1/C2/SP dens region

Insertion: Occiput / C1

Innervation: Suboccipital nerve C1 (dorsal ramus)

Function: Fine OA/AA control; proprioception; dens rotation (OCI)

Clinical Importance: Cervicogenic headache; muscle tension; CAD screen before aggressive MET

Exercises: Gentle nod control; avoid end-range rotation if CAD risk

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Sternocleidomastoid

ID: CERV-MUS-06

Muscle: Sternocleidomastoid

Origin: Manubrium & clavicle

Insertion: Mastoid / superior nuchal line

Innervation: Spinal accessory XI + C2–C3

Function: Ipsilateral SB, contralateral rotation; bilateral flexion

Clinical Importance: Torticollis; accessory nerve; scales with breathing; CAD relative contraindication caution for end-range

Exercises: Side-lying isometrics; avoid ballistic stretching

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Scalenes (Ant/Med/Post)

ID: CERV-MUS-07

Muscle: Scalenes (Ant/Med/Post)

Origin: Cervical TP

Insertion: 1st/2nd ribs

Innervation: C3–C8 ventral rami

Function: SB; elevate ribs; anterior contributes contralateral rotation assist

Clinical Importance: TOS thoracic outlet; respiratory accessory; brachial plexus relation

Exercises: 1st rib mobilizing breathing; scalene stretches careful; neural screen

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Levator Scapulae

ID: CERV-MUS-08

Muscle: Levator Scapulae

Origin: C1–C4 TP

Insertion: Superior medial scapular border

Innervation: Dorsal scapular C5 + C3–C4

Function: Elevate scapula; ipsilateral SB/rotation of neck

Clinical Importance: Trigger points to neck/scapula; with upper trap dominance

Exercises: Scapular upward rotation training (lower trap/serratus) to unload

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Upper Trapezius

ID: CERV-MUS-09

Muscle: Upper Trapezius

Origin: Occiput/nuchal/C7–T12 SP continuum upper fibers

Insertion: Clavicle/acromion

Innervation: XI + C3–C4

Function: Elevate/upwardly rotate scapula; ipsilateral SB contralateral rotation neck

Clinical Importance: Overactivity with neck pain; headache contributor

Exercises: Scapular setting; reduce shrugging; lower/mid trap emphasis

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Splenius Capitis/Cervicis

ID: CERV-MUS-10

Muscle: Splenius Capitis/Cervicis

Origin: Nuchal/C7–T6 SPs

Insertion: Mastoid/occiput; C1–C3 TP

Innervation: Dorsal rami

Function: Extension; ipsilateral SB/rotation

Clinical Importance: Global extensor overactivity vs deep stabilizer weakness

Exercises: Retrain deep before global heavy loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

5. Nerves

RECORD: Cervical Neural & Plexus Overview

ID: CERV-NRV-01

Dermatomes: C2 occiput/behind ear; C3 neck collar; C4 shoulder tip/upper trap; C5 lateral arm; C6 thumb/index lateral forearm; C7 middle finger; C8 little finger/medial hand; T1 medial forearm — classic teaching maps (overlap exists)

Myotomes: C4 diaphragm/shoulder shrug; C5 shoulder ABD; C6 elbow flex/wrist ext; C7 elbow ext/wrist flex; C8 finger flex; T1 intrinsics — screen bilaterally

Reflexes: Biceps C5–6; brachioradialis C6; triceps C7

Cranial_MSK_Relevance: CN XI (SCM/trapezius); CN V (TMJ/headache overlap); CN II–VIII in dizziness/vision differential for cervicogenic vs central

Brachial_Plexus: Roots C5–T1 → trunks/divisions/cords → terminal nerves (musculocutaneous, axillary, radial, median, ulnar). Scalene/1st rib/clavicle compression → TOS continuum

Radiculopathy_vs_Myelopathy: Root: unilateral arm pain/dermatomal/myotomal. Cord: gait, hand clumsiness, hyperreflexia, Babinski/Hoffmann, bowel/bladder — urgent imaging

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

6. Biomechanics

RECORD: Flexion

ID: CERV-BIO-01

Movement: Cervical flexion

Primary: Longus colli/capitis, SCM bilateral

Notes: Upper vs lower cervical differentiation important clinically

Clinical: Disc loading increases; myelopathy may worsen with extension more often but individualize

Exercises: CCFT, supine flexor endurance

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Extension

ID: CERV-BIO-02

Movement: Cervical extension

Primary: Semispinalis, splenius, multifidus, trapezius

Notes: Facet loading increases; foraminal volume decreases

Clinical: Extension-rotation may provoke CAD — screen per guidelines

Exercises: Deep extensor endurance prone graded

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Rotation

ID: CERV-BIO-03

Movement: Cervical rotation

Primary: AA complex major contributor; SCM contralateral; splenius ipsilateral

Notes: ~50% from C1–2 teaching model

Clinical: AA instability contraindication to aggressive rotation

Exercises: Controlled rotation in neutral with DNF

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Side Bending

ID: CERV-BIO-04

Movement: Side bending

Primary: Scalenes, levator, upper trap ipsilateral

Notes: Coupled with rotation in lower cervical (ipsilateral typically)

Clinical: Uncovertebral osteophytes limit SB

Exercises: Gentle SB with support

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Coupled Motion

ID: CERV-BIO-05

Movement: Coupled SB-rotation

Primary: Facet/uncinate guided

Notes: Lower cervical: SB and rotation same direction (classic); upper cervical opposite patterns often taught — simplify clinically with assessment

Clinical: Manual therapy coupling principles

Exercises: Combined movement control

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Deep Neck Stability

ID: CERV-BIO-06

Movement: Segmental stability / DNF–multifidus

Primary: Longus colli/capitis + multifidus

Notes: Feed-forward activation before arm load in healthy patterns

Clinical: Core of modern neck pain rehab (Jull et al. themes)

Exercises: CCFT biofeedback, co-contraction

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Gaze Stabilization

ID: CERV-BIO-07

Movement: Cervico-ocular / gaze

Primary: Suboccipitals + vestibulo-ocular systems

Notes: Neck afferents contribute to postural orientation

Clinical: Whiplash dizziness differential vs vestibular

Exercises: Gaze stability, smooth pursuit screen for central red flags

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Load Transfer / Posture / Sport

ID: CERV-BIO-08

Movement: Sitting, running, lifting head-neck load

Primary: Endurance of deep flexors/extensors

Notes: Forward head increases lever arm on extensors

Clinical: Ergonomics + capacity > 'perfect posture' alone

Exercises: Endurance holds, graded lifting with brace strategy

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

7. Pathologies

RECORD: Cervical Strain / Sprain

ID: CERV-PATH-01

Condition: Cervical strain/sprain

Mechanism: Overload/whiplash soft tissue

Symptoms: Pain, stiffness, muscle spasm

Red_Flags: Hard neuro signs, trauma imaging rules

Treatment: Early activity, advice, exercise; avoid prolonged collar usually

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Facet Syndrome

ID: CERV-PATH-02

Condition: Cervical facet syndrome

Mechanism: Extension-rotation overload; arthropathy

Symptoms: Localized neck pain; referral per Bogduk maps; headache if upper

Red_Flags: Myelopathy signs

Treatment: Manual therapy + exercise; interventional dual blocks/RF selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Disc Herniation / Radiculopathy

ID: CERV-PATH-03

Condition: Cervical disc herniation with radiculopathy

Mechanism: Posterolateral disc → root

Symptoms: Neck/arm pain, dermatomal sensory, myotomal weakness, reflex change

Red_Flags: Progressive deficit, myelopathy

Treatment: Majority improve nonsurgically; PT, meds; ACDF/foraminotomy selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Cervical Myelopathy

ID: CERV-PATH-04

Condition: Cervical spondylotic myelopathy

Mechanism: Cord compression — stenosis/OPLL/disc

Symptoms: Clumsy hands, gait disturbance, hyperreflexia, Hoffmann/Babinski, bladder changes

Red_Flags: THIS IS A RED FLAG — urgent surgical opinion

Treatment: Decompression often; do not mobilize aggressively

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Whiplash (WAD)

ID: CERV-PATH-05

Condition: Whiplash-associated disorder

Mechanism: Acceleration-deceleration energy transfer to neck

Symptoms: Pain, stiffness, headache, dizziness, cognitive complaints variable

Red_Flags: Canadian C-Spine/NEXUS; neurologic deficit; CAD features

Treatment: Early activation, education, exercise; grade-based care (QTF)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Cervicogenic Headache

ID: CERV-PATH-06

Condition: Cervicogenic headache

Mechanism: Upper cervical afferent convergence (C1–C3)

Symptoms: Unilateral neck-triggered headache, limited ROM, manual palpation provocation

Red_Flags: SNNOOP10 headache red flags — sudden/ thunderclap, neuro deficit, etc.

Treatment: Manual therapy + DNF/exercise (evidence supportive)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Thoracic Outlet Syndrome

ID: CERV-PATH-07

Condition: TOS (neurogenic most common)

Mechanism: Brachial plexus ± vessel compression at scalene/1st rib/pec minor

Symptoms: Arm pain/paresthesia with overhead; vascular signs if arterial/venous

Red_Flags: Acute ischemia / effort thrombosis — urgent

Treatment: Scapular/postural PT first for NTOS; surgery selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

RECORD: Degeneration / Stenosis

ID: CERV-PATH-08

Condition: Spondylosis and stenosis

Mechanism: Age-related disc/facet/uncovertebral hypertrophy

Symptoms: Stiffness, foraminal or central stenosis syndromes

Red_Flags: Myelopathy

Treatment: Exercise, traction trial, surgery for cord/progressive root

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes.

8. Rehabilitation

RECORD: Cervical Rehabilitation Principles

ID: CERV-REHAB-01

Acute: Screen red flags/CAD/imaging rules; advice to remain active; short-term pain strategies

Motor_Control: Cranio-cervical flexion training; multifidus co-contraction; scapular mechanics

Mobility: Restore rotation/SB/extension as appropriate directional preference

Strength_Endurance: Deep flexor/extensor endurance; axioscapular strength

Sensorimotor: Joint position error, gaze stability if dizziness (after central screen)

Functional: Ergonomics, lifting, sport return graded

Outcome_Measures: NDI, NPRS, PSFS, headache diary

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bogduk N. Clinical Anatomy of the Cervical Spine. Magee DJ. APTA Neck Pain CPG themes. APTA Neck Pain CPG; Jull deep cervical flexor research themes.